MIL-P-12951F (AR)
9 August 1972
SUPERSEDING
MIL-P-12951E (AR)
9 October 1969

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M39A1 PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing of one type of percussion primer designated as Primer, Percussion M39A1

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8797919	–	Primer, Percussion, M39A1 Assembly
8864583	–	Container, Ammunition, Plastic for Percussion Primer, M39A1
8864584	–	Box, Wire Bound, Packing, Ammunition for Percussion Primer M39A1
9207697	–	Carton, Packing, Ammunition, for Percussion Primer, M39A1 Level C Packing

PUBLICATIONS

U.S. ARMY MUNITIONS COMMAND

ET-8798919 – Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.- Material and parts shall be in accordance with applicable drawings and specifications.

3.2 Primers.-The primers shall comply with all requirements specified on Drawing (dwg.) 8798919, and with all requirements specified in applicable specifications.

3.3 Moisture content of primer mixture composition.-The moisture content at the loading station at the time of loading the primer shall not exceed 0.30 percent when determined as specified in 4.4.1.

3.4 Functioning.-The primer assembly shall not show evidence of a firing failure (see 6.5) when tested as follows:

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3.4.1 Sensitivity.-The primer assembly shall fire without any evidence of failure when a maximum (max.) of twenty-nine and forty hundredths (29.40) inch ounces (see 6.6) of energy is applied and shall not fire when a minimum (min.) of three and eighty-four hundredths (3.84) inch ounces of energy is applied. The test shall be performed as specified in 4.4.2.1.

3.4.2 Failure to fire (Dud).-The primer assembly shall not fail to fire when tested as specified in 4.4.2.2.

3.5 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. All packing components shall be dry.

3.6 First article inspection.-This specification makes provision for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a statement of findings on the moisture content of primer mixture composition shall be furnished with each lot submitted for acceptance.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five (25) each of anvils, cups, and bodies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including certificates of conformance of materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

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4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an item to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded primer assemblies shall contain:

a. Like parts from not more than one supplier.

b. Primer mixture composition from not more than one batch or blend of batches which results in homogeneity of all primer mixture used in loading a lot.

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4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Body (see dwg. 8798921 covering a detail of dwg. 8798919)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Inside diameter	Gage	01001
102.	Outside diameter, maximum (max). open end	Gage	01002
Minor:	AQL 4.00 percent		
201.	Total length, max	Gage	01003
202.	Thickness through bottom, max	Gage	01004
203.	Radius incorrect or missing	Visual	01005
204.	Finish improper	Visual	01006
205.	Marking missing or incorrect	Visual	01007
206.	Evidence of poor workmanship, (see 3.5)	Visual	01008

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4.3.2.2 Cup (see dwg. 8798922 covering a detail of dwg. 8798919).

Categories	Defects	Method of Inspection	Code No
Critical:			
1.	Crack or split	Visual	02001
Major:	AQL 0.65 percent		
101.	Outside diameter, max	Gage	02002
102.	Thickness through bottom, max	Gage	02003
103.	Total length	Gage	02004
104.	Shellac missing or incomplete	Visual	02005
Minor:	AQL 1.00 percent		
201.	Inside diameter	Gage	02006
202.	Evidence of poor workmanship, (see 3.5)	Visual	02007

4.3.2.3 Anvil (see dwg. 8798920 covering a detail of dwg. 8798919).

Categories	Defects	Method of Inspection	Code No
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Distance from rear to shoulder	Gage	03001
102.	Total length	Gage	03002
103.	Striker radius	Gage	03003
Minor:	AQL 2.50 percent		
201.	Width under shoulders, max	Gage	03004
202.	Total width, max	Gage	03005
203.	Thickness	Gage	03006
204.	Radii missing at rear end	Visual	03007
205.	Evidence of poor workmanship, (see 3.5)	Visual	03008

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4.3.2.4 Primer assembly, prior to assembly of anvil (see dwg. 8798919).

Categories	Defects	Method of Inspection	Code No
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Charge missing	Visual	04001
102.	Excessive number of discs	Visual	04002
103.	Foreign matter in cup	Visual	04003
104.	Disc missing	Visual	04004
Minor:	AQL 1.00 percent		
201.	Disc damaged or improperly inserted	Visual	04005
202.	Evidence of poor workmanship, (see 3.5)	Visual	04006

4.3.2.5 Primer assembly (see dwg. 8798919)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Outside diameter, max	Gage	05001
102.	Total length, max	Gage	05002
103.	Anvil missing, inverted or loose	Visual	05003
104.	Body cracked, dented or distorted	Visual	05004
105.	Cocked anvil	Visual	05005
106.	Vent hole missing	Visual	05006
107.	Hole diameter, minimum (min.)	Gage	05007
Minor:	AQL 1.50 percent		
201.	Thickness through flange, max	Gage	05008
202.	Diameter of flange, max	Gage	05009
203.	Crimp incomplete	Visual	05010
204.	Marking missing or incorrect	Visual	05011
205.	Evidence of poor workmanship, (see 3.5)	Visual	05012

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4.3.2.6 Container or set-up box, prior to sealing (see dwg. 8864583 or 9207697)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Tray missing or damaged	Visual	06001
102.	Primer missing	Visual	06002
Minor:	AQL 1.00 percent		
201.	Packing component missing	Visual	06003
202.	Evidence of poor workmanship, (see 3.5)	Visual	06004

4.3.2.7 Sealed plastic container, set-up box and carton (see dwg. 8864583 and 9207697).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Sealing tape incomplete or badly wrinkled	Visual	07001
102.	Container cut, punctured or torn	Visual	07002
Minor:	AQL 1.00 percent		
201.	Marking missing, incorrect or illegible	Visual	07003
202.	Evidence of poor workmanship, (see 3.5)	Visual	07004

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4.3.2.8 Packing box (see dwg. 8864584).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Board broken or split	Visual	08001
102.	Hardware or wire missing, broken or loose	Visual	08002
Minor:	AQL 1.50 percent		
201.	Hardware or wire improperly engaged or mislocated	Visual	08003
202.	Metallic seal missing, unsealed or improperly positioned	Visual	08004
203.	Marking missing, incorrect or illegible	Visual	08005

4.3.3 Testing

4.3.3.1 Moisture content of primer mixture composition (see 3.3) - Major defect - Code No. 09001.-The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification one sample of the primer mixture composition from each eight hours production of primers. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If primers have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all primers loaded with the non-conforming material shall be rejected. The tests shall be performed as specified in 4.4.1.

4.3.3.2 Functioning

4.3.3.2.1 Sensitivity (see 3.4.1) - Major defect - Code No. 05013.-A sample of 700 percussion primers shall be selected from each lot for this test. If the sum of the average critical height (H average) plus 5 times the standard deviation (s) multiplied by the weight of the ball exceeds the maximum requirement, or if the sum of the critical height (H average) minus 2 times the standard deviation (s) multiplied by the weight of the ball is less than the min. requirement the lot shall be rejected (see 6.7). The test shall be performed as specified in 4.4.2.1 using equipment in accordance with 4.3.4.

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4.3.3.2.2 Duds (see 3.4.2) - Major defect - Code No. 05014.-Three hundred (300) primer assemblies shall be selected from each lot for this test. If two (2) or more primers fails to comply with the requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2.2 using equipment in accordance with 4.3.4.

4.3.3.3 Composition of primer mixture composition (see dwg. 8798919) - Major defect - Code No. 09002.-The composition of each batch or blend of batches of primer mixture composition shall be determined at the time of loading. A sample of sufficient size to perform the test specified in 4.4.3 shall be selected. If the composition of the sample fails to comply with the requirement, the quantity of primer mixture represented by the sample shall be rejected. Any primer assemblies loaded with the non-conforming composition shall also be rejected.

4.3.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified by Equipment Tabulation ET-8798919.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Moisture content of primer mixture composition.-Transfer a known weight of the sample to a tared weighing dish approximately 55 millimeters (mm) in diameter and 30 mm in height. Place the dish and contents in a desiccator containing anhydrous calcium chloride until constant weight is obtained. Calculate the loss in weight as the percentage of moisture.

4.2.2 Functioning.-The primer assemblies shall be conditioned at a temperature of seventy-five (75) plus or minus five (5) degrees Fahrenheit (F.) for twenty-four (24) hours min. prior to performing the following tests.

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4.4.2.1 Sensitivity.-The sample of 700 primer assemblies shall be divided into groups of 50 primers each. A group shall be selected, and each primer in the group shall be assembled in the test fixture specified in 4.3.4, and tested by dropping a steel ball weighing 1.94 plus or minus .02 ounces on the firing pin from a height of 8 inches. The drop height is measured as the distance from the bottom of the ball to the top of the firing pin. The number of primers that fire, or exhibit a firing failure shall be recorded. This procedure shall be repeated at successively increasing heights of one inch using a new group of 50 primers at each height until a height is reached at which all primers fire in the group being tested. The test fixture shall then be set at a height of 7 inches and the above procedure repeated at successively decreasing heights of one inch, using a new group of 50 primers at each height, until a height is reached at which all primers in the group being tested do not fire. Acceptance or rejection of the lot shall be determined from calculations based on recorded data (see 6.7). All samples subjected to test shall be scrapped, but untested samples may be returned to the lot.

4.4.2.2 Duds.-Each primer shall be assembled in the test fixture specified in 4.3.4 and a steel ball weighing 1.94 plus or minus .02 ounces dropped on the firing pin from a height of 15 inches max. The number of primers that exhibit a firing failure shall be recorded to determine compliance with the requirement.

4.4.3 Determination of primer mixture composition

4.4.3.1 Trinitrotoluene (TNT).-To an accurately weighed sample of about 0.5 gram in a 150 milliliter (ml.) beaker, 25 ml. of ethyl ether shall be added and stirred to dissolve the TNT. The solution shall be filtered through a tared filtering crucible, transferring the insoluble residue with small portions of ether. The crucible and contents shall be washed with ether, dried in an oven maintained at 100 degrees centigrade (C) plus or minus 5 degrees C., cooled and weighed. The loss in weight shall be calculated as the percentage of TNT.

4.4.3.2 Ground glass.-The material insoluble in ether shall be washed with 25 ml. of 5 percent potassium nitrate solution in 5 ml. portions. Each portion shall be allowed to remain in contact with the residue for 2 to 3 minutes before using vacuum. The material shall be washed with water at room temperature, dried in an oven maintained at 100 plus or minus 5 degrees C., cooled and weighed. The weight of the insoluble material shall be calculated as the percentage of ground glass.

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4.4.3.3 Lead sulphocyanate.-To combined filtrate and washings from the determination of ground glass, an excess of N/10 silver nitrate solution shall be added. The solution shall be boiled, the precipitate shall be allowed to settle, and then the solution filtered through a tared filtering crucible. The crucible and contents shall be washed with water, dried in an oven maintained at 135 plus or minus 5 degrees C., cooled and weighed. The weight of precipitate shall be calculated as the percentage of lead sulphocyanate as follows:

Lead sulphocyanate = 0.9742 X silver sulphocyanate

4.4.3.4 Barium nitrate.-The filtrate and washings from the determination of lead sulphocyanate shall be evaporated to about 100 ml. To the boiling solution 10 ml. of 10 percent sulphuric acid shall be added. When the precipitate has settled, the mixture shall be filtered through a tared, previously ignited filtering crucible, washed with four 10 ml. portions of acid ammonium acetate (120 ml. of concentrated acetic acid, 100 ml. of distilled water, and 96 ml. of concentrated ammonium hydroxide) washed with distilled water, and then the precipitate shall be ignited to a dull red heat. The crucible and contents shall be allowed to cool and then weighed. The percentage of barium nitrate in the sample shall be calculated as follows:

Barium nitrate = 1.1197 X barium sulfate

4.4.3.5 Potassium chlorate.-Calculate the percentage of potassium chlorate by subtracting from 100 percent the sum of the percentages of TNT, ground glass, lead sulphocyanate and barium nitrate.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The primer assemblies shall be preserved and packaged in accordance with dwg. 8864583.

5.1.2 Level C.-The primer assemblies shall be preserved and packaged in accordance with dwg. 9207697.

5.2 Packing

5.2.1 Level A.-The containers shall be packed in accordance with dwg. 8864584.

5.2.2 Level C.-The packages shall be packed in accordance with dwg. 9207697.

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5.3 Marking.-Unit and exterior containers shall be marked in accordance with instructions contained in dwgs. 8864583, 8864584 and 9207697 as applicable.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers on the Equipment Lists referenced on the Equipment Tabulation. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

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6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204 "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (for other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-M, Dover, New Jersey, 07801 in accordance with the stipulations, time frame and distribution specified in the contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of ammunition data cards.-In addition to the normal distribution of records, one copy of all ammunition data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SMUPA-QA-A-M
(2) SMUPA-ADEP-2

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6.5 Firing failure.-A firing failure is defined as follows:

a. Dud.-A failure wherein initiation of the primer mixture does not occur.

b. Squib.-A failure wherein the primer mixture burns with a hiss or fizz.

c. Hangfire.-A failure wherein an audible delay occurs between the application of energy and the detonation report.

6.6 Inch ounces.-The product of drop height in inches multiplied by the weight of the ball in ounces.

6.7 Formula and method of determining H average and s.

6.7.1 Formula

H average = A plus B plus C
s = I times the square root of L minus M

where: (see 6.7.2)
A = the sum of column 3
B = the first drop height at which all primers misfired
C = One-half the drop height increment (see 6.8)
L = the sum of column 5
M = the square of the sum of column 3
I - the drop height increment

6.7.2 Tabulation of data.-Divide a sheet of paper into 5 columns and tabulate the data as follows:

a. Title Column I "Height of Drop" and list all drop heights in order starting with the lowest height. DO NOT list the heights at which all elements fired or misfired.

b. Title Column 2 "Number of Failures" and list the sum of the number of elements that misfired, squibbed or hangfire (see 6.5) at each intermediate height.

c. Title Column 3 "Decimal Fraction of Elements Failing to Fire at Each Intermediate Height" and record the decimal fraction that failed to fire at each intermediate height. This number is obtained by dividing the value obtained under "Number of Failures" by the number of elements tested at that height. Caculation should be carried out and recorded to the nearest one-hundredth (0.01).

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d. Title Column 4 "Standard Deviation Factor" and list odd numerals in sequence starting with 1, for as many drop heights as are listed in Column 1.

e. Title Column 5 "Standard Deivation Factor Times Fraction of Failures" and, by multiplying each decimal fraction in Column 3 by the number on the same line in Column 4, enter the product on the corresponding line in Column 5.

6.7.3 Procedure for solving equations:

a. To obtain "A" - Add the values in Column 3 and substitute the sum for "A".

b. To obtain "B" - The first drop height at which all primers failed to fire.

c. To obtain "C" - Divide the drop height increment by 2 and substitute the dividend for "C".

d. To obtain "L" - Add the value in Column 5 and substitute the sum for "L".

e. To obtain "M" - Square the sum of Column 3 and substitute the sum for "M".

f. To obtain "H average" - Add the values substituted for A, B and C.

g. To obtain "s" - Subtract the value substituted for "M" from the value substituted for "L" and then obtain the square root of the difference and then multiply the square root by the height of increment.

h. Using the calculated values for "H average" plus 5s mulitplied by the maximum weight of the ball is equal to or less than the maximum inch ounce energy value required, and "H average" minus 2s multiplied by the minimum weighted of the ball is equal to or greater than the minimum energy value required, the lot should be considered acceptable for sensitivity.

6.8 Height of increment.-The height of increment is defined as a constant interval in inches between successive drop heights which must be considered in calculating average height and the standard deviation in drop height, whenever it differs from one inch.

6.9 Check test for possible deterioration.-The following test is designed to prevent major ammunition items being loaded with primers that may have deteriorated. The test should be performed on primers from any lot which has been

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stored for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are loaded does not require that the check test be performed the contracting officer should insert provisions for the performance of the check test in the contract. The contracting officer should arrange to have the contractor reimbursed for any expenses incurred in performance of the check test for possible deterioration.

6.9.1 Sensitivity.-The test shall be performed as specified in 4.4.2.1 using test equipment specified in 4.3.4. A sample of 1050 primers shall be selected from each lot for this test. Seventy-five primers shall be tested at each height. If H average plus 4.66s exceeds the upper limit specified in 3.4.1, or if H average minus 1.86s is less than the lower limit specified in 3.4.1 the lot shall be rejected.

6.9.2 Duds.-This test shall be performed as specified in 4.4.2.2. A sample of 300 primers shall be selected from each lot for this test. Failure to function of 3 or more primers shall reject the lot.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1375-A-020

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